MA685 Project 1 Requirements – Due: [Friday 11/6/26] (UL submission at 11:59pm; no late submissions) – Star Count: [3] (tier based)

Optional (partial-)draft submission due [Monday 11/2/26] (printed in person in class) if you want feedback on structure.

Step 1: Obtain dataset(s) that allow the group to conduct the analyses in the Project Sections below.

- The dataset(s) must be legitimate, not artificially generated, and must be obtainable from the public (usually via a hyperlink or GitHub). Possible locations to download are Kaggle, USA Census, US.gov, etc.
- Dataset(s) must contain at least 300 observations with at least 2 categorical variables and at least 4 numerical variables. At least one categorical variable must be binary (for the binomial logistic regression), and at least one must have 3 or more groups (for the multinomial logistic regression and the LDA).
- If you happen to use more than one dataset, the theme should be the same across all datasets. For example, all should be sports related, politics related, gaming related, and so on, with the questions and discussion all circling around one main idea.
- The group leader must upload the dataset(s) used for the project (the exact file(s) the analysis was run on, such as a .csv) to UL along with the project submission, so that every result in the report can be reproduced from it.

Step 2: State the prediction and classification questions (in words) associated to your project theme that the models in the Project Sections will be used to answer.

Writing Format: The project must be done either in LaTeX (such as in Overleaf) or in R-Markdown with LaTeX-based equations. The project should be in a standard discussion and research format, where all questions, calculations, and methodology are discussed in a conversational and logical manner. All code used to generate any result in the project must be included in an appendix at the end of the paper, presented in a professional manner; code may also appear in the body of the paper where it helps the discussion. All functions called that you wrote must be included in the appendix as well (that is, no "black-box" function calls/sourcing). No external libraries/packages are permitted outside of the ones used in lecture (such as glm(...) for binomial logistic regression and multinom(...) from nnet for multinomial logistic regression). The paper must have a title page, table of contents, and bibliography. All graphs/figures must be properly labeled, center justified, and not take up an overwhelming amount of space. Pages in your report that contain text and/or figures that span 20% or less of the page should be reformatted such that such "empty-space" pages do not exist in your report.

Project Sections:

1. **Introduction**: Discusses the problems solved in the submission, discusses the real-world implications of the conclusions for which those problem solutions reveal, and also discusses potential drawbacks or illusions for which the dataset will contribute to such problems, solutions, and implications (for example, issues tied to data collection).
2. **Binomial & Multinomial Logistic Regression for Predicting Likelihoods**:
    a. Must fit at least one binomial logistic regression model and at least one multinomial logistic regression model.
    b. The function which is optimized to generate the coefficients (the log-likelihood) should be presented in LaTeX for both models.
    c. Model performance metrics (at least 3) should be included and discussed, such as RMSE, AIC, BIC, and the McFadden pseudo-R^2.
3. **Classification with Logistic & Multinomial Logistic Regression**: You can use the classification as a follow up to the previous section (if appropriate), or a separate problem/dataset if desired. Results and consequences (in addition to discussion) should be included.
    a. Binary classifications should be made with a stated threshold, and multinomial classifications via the maximum predicted probability.
    b. Classification accuracy should be reported on data the model was not built with.

4. **Testing/Training/Hyper-Parameter Tuning**: You can apply all of these concepts to one model, or multiple models (whichever feels appropriate). Methodology and discussion of computations should be given in detail, and results should be clearly presented, both qualitatively and quantitatively.
    a. The threshold (or other hyper-parameter) chosen must be justified over more than one random testing/training partition (aggregation over seeds), with a discussion of how much the partition affects the result.
    b. Variable selection (backward elimination or a comparison of all models) should be used to choose the predictors of at least one model; variable selection procedures cannot be used to demonstrate testing, training, or tuning.
5. **Dimensionality Reduction**: Should use PCA and also LDA in two separate but appropriate scenarios. Ideally these should be separate from the items above, but can be intertwined if appropriate (I don't recommend it, since it will affect the coefficient interpretation of the models above). Both methods should be used as precursors for different regression/classification models.
    a. The number of components kept must be justified (for example, with the Kaiser criterion, the elbow rule, and/or an explained-variance threshold).
    b. Discussions associated to significant (or lack thereof) component loadings should at the least be mentioned for interpretation of reduced variables.
6. **Conclusion and Discussion of Results**: Should be self explanatory; should reference each major section of your submission.

Miscellaneous Requirements:

- Member Contribution: Each team member should be listed (first and last name), in addition to what they contributed to the entire submission. Roughly uniform distribution of contribution is expected.
- All items mentioned above should be motivated (with an opening for each section) and discussed throughout your calculations. Discuss the potential impact of the results, as well as the consequences the results may have in the case that the assumptions of the models are not met.
- RMSE, accuracy, AIC, BIC, McFadden pseudo-R^2, Gini impurity, etc. are not really a separate topic, but should be naturally intertwined in all items mentioned above in an appropriate manner.
- There must be at least one appropriate use of each of: (a) bootstrapping, (b) a Monte-Carlo simulation, (c) Gini impurity, and (d) a log-likelihood (or other objective function) optimized numerically with optim(...), compared against the corresponding glm(...) or multinom(...) result.
- There must be at least three appropriate uses of graphical representations (using base R graphing techniques), including but not limited to histograms and scree plots.
- There must be at least six appropriate and meaningful uses of LaTeX code for displayed equations, not including in-line/in-paragraph equations.